Sales Decline Diagnosis: Volume, Mix & Margin Analysis

This project investigates a decline in sales performance at a U.S.-based multi-category retail company across Electronics, Furniture, and Office Supplies. In 2025, the company recorded a 9.4% YoY decrease in total sales, despite relatively stable profit margins, indicating that the issue is likely driven by weakened revenue generation rather than cost inefficiencies.

This analysis aimed to:

Identify the key drivers of this decline by examining performance across categories and sub-categories and to generate actionable insights that support data-driven strategies to restore sustainable growth.

Business Questions

- Is the 2025 sales decline driven by lower transaction volume, price changes, or product mix shifts?
- Which categories and sub-categories contribute to growth versus eroding value?
- Is profitability truly resilient, or are healthy margins masking underlying volume- or mix-driven risks?
- Which segments should be prioritized to safeguard profitability and restore sustainable growth?

Key Findings

- **Sales decline is volume-driven, not price-driven.** The 9.4% YoY decline was primarily driven by a 6.2% drop in quantity sold, while AOV remained stable, indicating weakening demand rather than changes in pricing or transaction value.
- **Demand is structurally unstable, not seasonal.** With 8 of 12 months below the annual average, the pattern reflects persistent weakness rather than seasonal fluctuation.
- **Growth is highly concentrated in Electronics.** Electronics (+18.7%) is the sole growth driver, while Office Supplies (-25.4%) and Furniture (-15.8%) are dragging overall sales performance.
- **Profitability remains strong but potentially misleading.** Margins rose (+7.6% YoY) despite declining sales, signaling gains driven by pricing or product mix rather than underlying demand. In Office Supplies, higher margins alongside sharp volume declines point to a risk of amplifying demand contraction.
- **Office Supplies is the most structurally challenged category.** Broad volume declines, high volatility, and margin expansion masking demand weakness suggest continued downside risk without targeted intervention.

Deep Dive Analysis

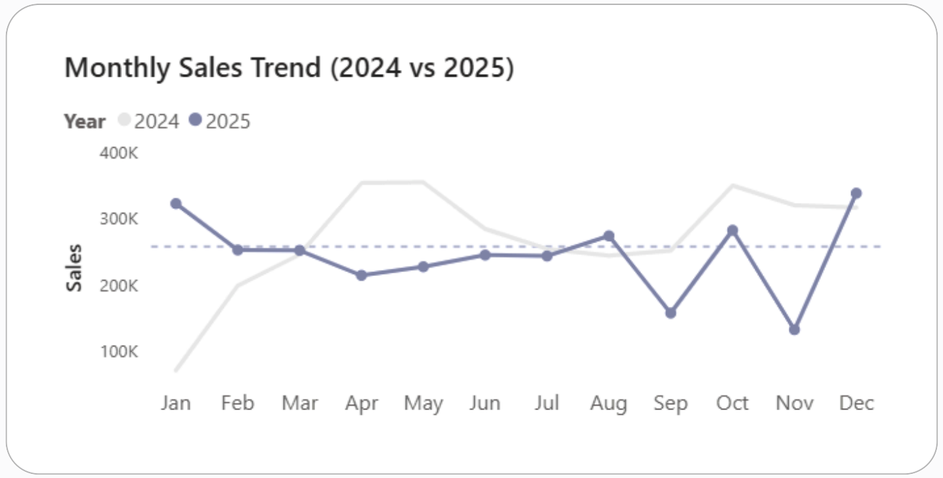
KPI Overview

SALES	QUANTITY	AOV	PROFIT	PROFIT MARGIN
$2.94M	6169	9.33K	795K	27%
vs Last Year ▼ -9.4%	vs Last Year ▼ -6.2%	vs Last Year ▲ 0.4%	vs Last Year ▼ -2.5%	vs Last Year ▲ 7.6%

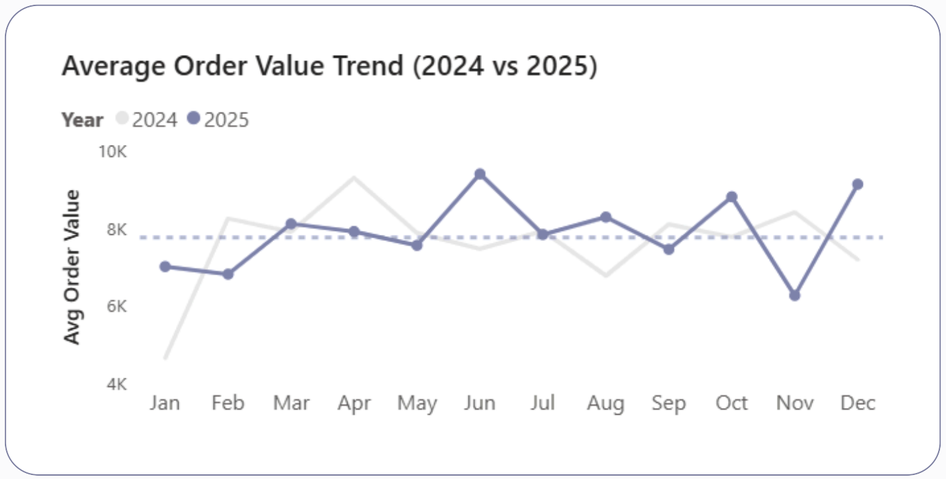
Sales declined 9.4% YoY to **$2.94M,** driven by a **6.2% drop in quantity**, while AOV remained stable (+0.4%), indicating consistent spending per transaction. Profit fell slightly (-2.5%), but **margin expanded to 27% (+7.6%)**, suggesting gains are likely driven by pricing or product mix shifts rather than underlying demand.

However, this margin improvement warrants caution: **rising profitability alongside declining volume suggests a trade-off between margin and scale.**

Sales Performance Overview

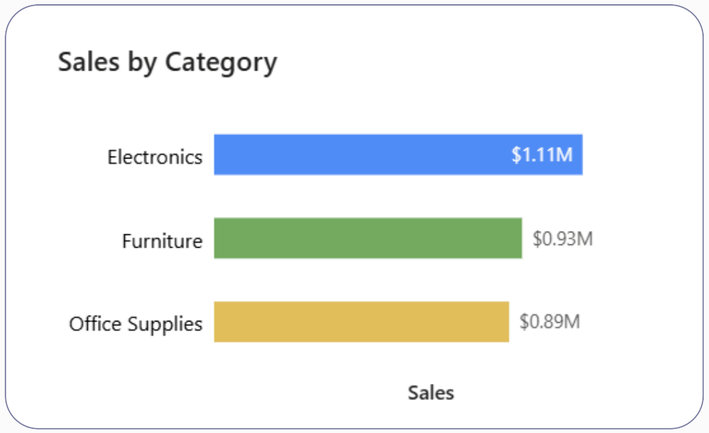


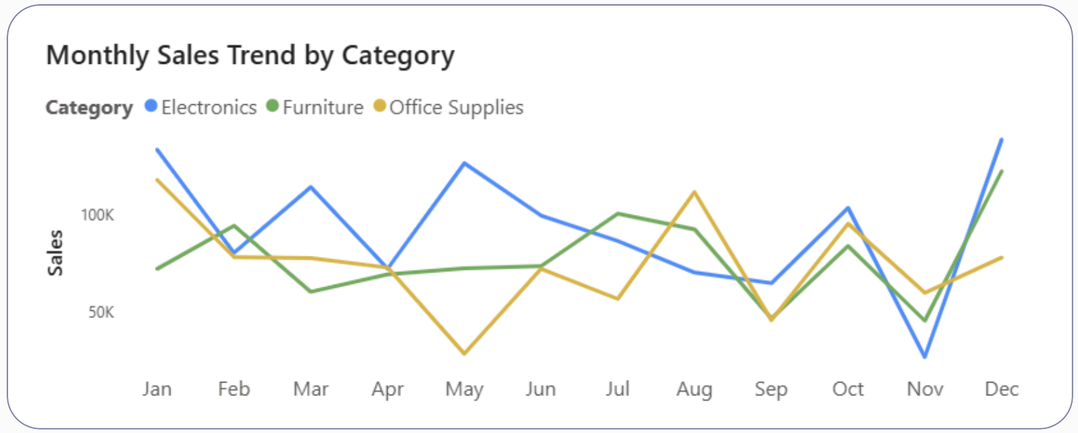
Monthly trends reveal **higher volatility in 2025 vs. 2024**, with **eight months falling below the annual average** and sharp declines in September and November, pointing to **unstable demand rather than a one-off seasonal effect**.



Despite this, **AOV showed no significant deterioration in 2025**, indicating that customers who transacted continued to spend at relatively consistent levels, confirming the decline was driven by **fewer transactions**, not lower basket value.

Sales Performance Overview



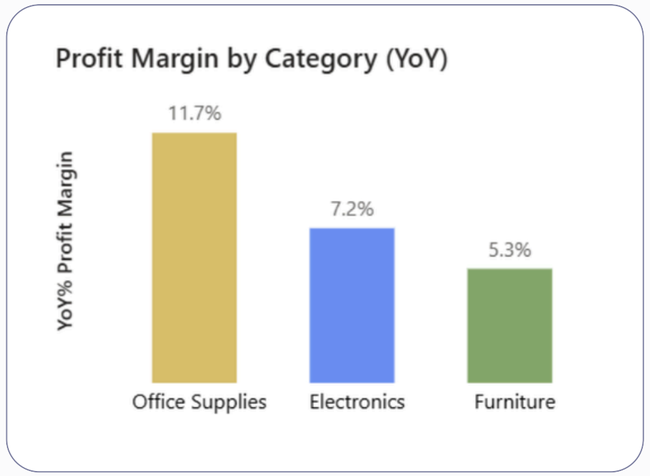


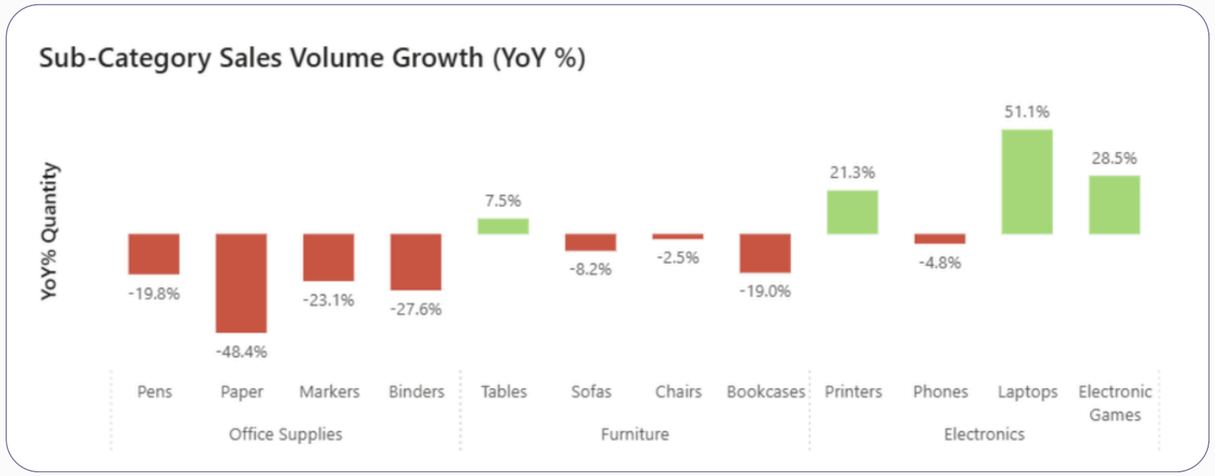
The 2025 sales contraction is highly concentrated at the category level. While **Office Supplies and Furniture experienced material demand contraction**, **Electronics remains the only category with sustained growth**.

Monthly category trends show diverging demand dynamics:

- **Electronics** displays stronger peaks and a clear December rebound, indicating resilient and seasonally-driven demand.
- **Office Supplies** is the most volatile category, with a pronounced mid-year trough, pointing to structural demand weakness.
- **Furniture** remains the most stable category, showing limited upside and weak recovery signals.

Sales Performance Overview



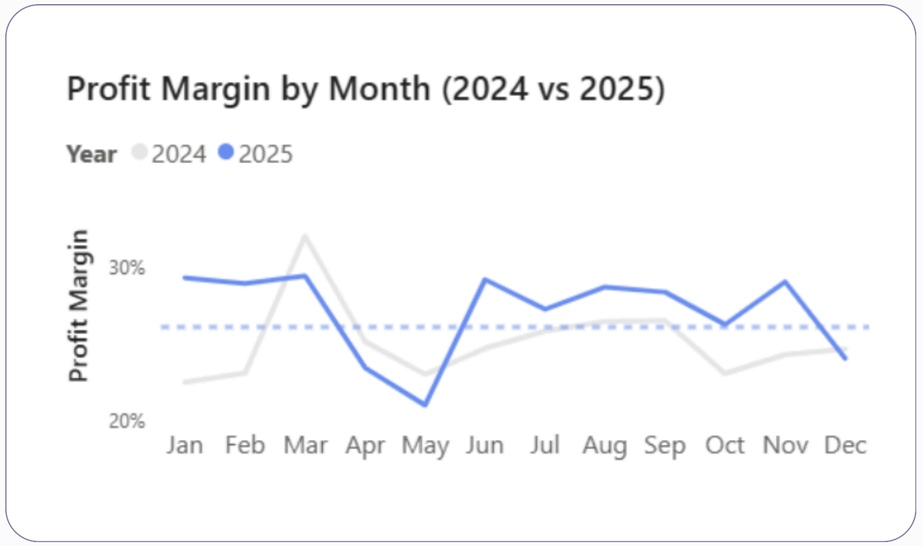


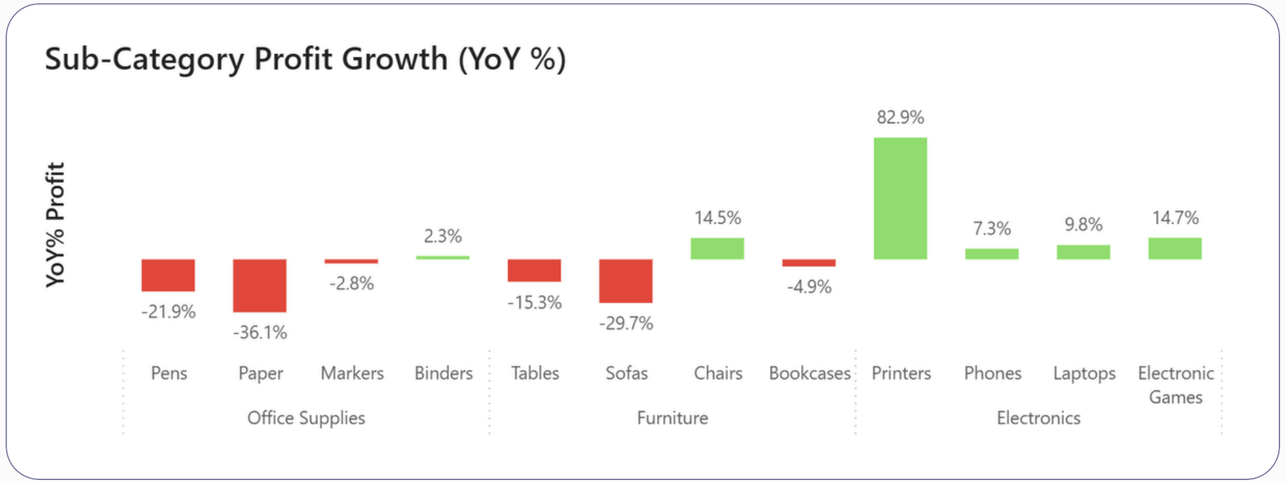
To understand these category-level dynamics, we examine performance at the sub-category level.

- **Office Supplies faces broad-based volume erosion**, particularly in **Paper (-48.4%)** and **Binders (-27.6%)**.
- **Furniture performance is mixed**, with **Tables (+7.5%)** as the only notable positive contributor.
- **Electronics growth is structurally driven**, led by **Laptops (+51.1%)** and **Electronic Games (+28.5%)**.

Margins improved across all categories (**+5–12% YoY**). However, **in Office Supplies, margin expansion alongside sharp volume decline**, suggesting **price sensitivity** and **mix effects** may be amplifying **demand loss rather than stabilizing the category**.

Sales Performance Overview





Overall margins remained stable between **22% and 30%** throughout the year, with **several months in 2025 outperforming 2024**, indicating **improved efficiency or pricing discipline**.

At the sub-category level:

- **Printers** (**+82.9% YoY profit**) and **Electronic Games** (**+14.7%**) show strong profit momentum.
- **Paper** (**-36%**) and **Pens** (**-22%**) experienced sharp profit declines, consistent with structural volume erosion.

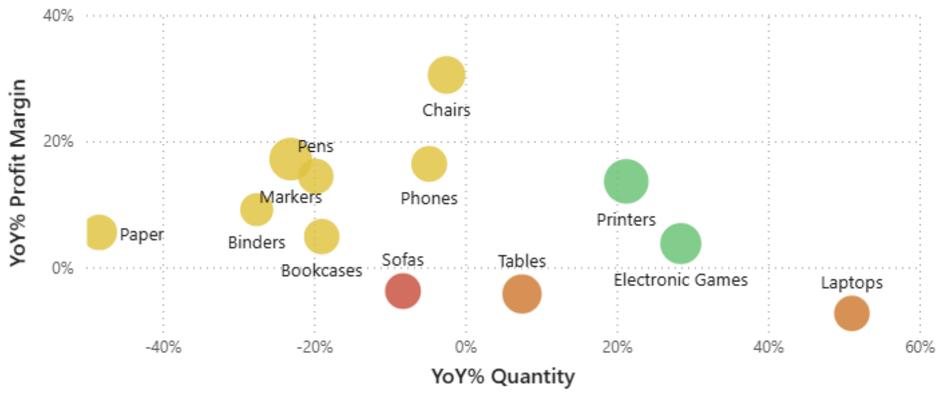
Sales Performance Overview

Sub-Category Performance Summary

Sub-Category	Profit CY	Margin CY	YoY% Margin	YoY% Quantity	Status
Electronic Games	79K	27%	3.8%	28.5%	Accelerate
Printers	95K	27%	13.6%	21.3%	Accelerate
Binders	49K	26%	9.2%	-27.6%	Defend
Bookcases	58K	29%	4.9%	-19.0%	Defend
Chairs	66K	32%	30.5%	-2.5%	Defend
Markers	86K	30%	17.2%	-23.1%	Defend
Paper	58K	30%	5.6%	-48.4%	Defend
Pens	57K	25%	14.4%	-19.8%	Defend
Phones	59K	24%	16.4%	-4.8%	Defend
Laptops	58K	25%	-7.3%	51.1%	Fix Pricing
Tables	72K	25%	-4.2%	7.5%	Fix Pricing
Sofas	59K	25%	-3.8%	-8.2%	Deprioritize

Volume vs. Margin by Sub-Category (Size by Profit)

Growth defined as YoY Volume ≥ 5% and Margin improvement ≥ 2 ppt



Sub-categories fall into **four strategic groups** based on volume and margin dynamics, with bubble size reflecting profit contribution:

- **Accelerate — Printers and Electronic Games** stand out as high-quality growth candidates, combining volume expansion with margin improvement.
- **Defend — Chairs, Pens, Markers, Paper, Bookcases, and Phones** maintain acceptable margins but face declining volumes, requiring selective defense rather than broad investment.
- **Fix Pricing — Laptops and Tables** show positive volume momentum but weaker margins, pointing to pricing or cost inefficiencies rather than demand issues.
- **Deprioritize — Sofas** show declining margins and volumes, suggesting limited value contribution.

Recommendations

Strategic Recommendations

1. Accelerate — Invest to Scale High-Quality Growth	
Product	Printers, Electronic Games
Insight	These products demonstrate quality growth, expanding volumes without sacrificing profitability making them the strongest candidates for incremental investment.
Objective	Maximize value creation by scaling segments that deliver sustainable, profitable growth.
Recommendations	• Prioritize capital, inventory, and marketing spend toward these sub-categories. • Protect margin integrity by avoiding unnecessary discounting; growth is demand-led, not price-led. • Invest in product innovation, feature differentiation, and targeted promotions to reinforce competitive advantage. • Strengthen supply chain reliability to support scaling without margin dilution.

Strategic Recommendations

2. Defend — Stabilize Volume Without Sacrificing Profitability	
Product	Chairs, Pens, Markers, Paper, Bookcases, Phones
Insight	These segments remain profitable but face structural or competitive demand pressure. The goal is value protection, not aggressive growth.
Objective	Prevent further volume erosion while preserving profitability.
Recommendations	• Diagnose root causes of volume decline (price sensitivity, substitution, channel shift, or demand saturation). • Deploy selective, targeted growth levers (bundling, contract pricing, loyalty programs) rather than broad discounts. • Maintain strict margin discipline; accept controlled volume decline where recovery would require margin-destructive pricing. • Rationalize low-velocity SKUs to reduce operational complexity and cost-to-serve.

Strategic Recommendations

3. Fix Pricing — Repair Margins Before Scale	
Product	Laptops, Tables
Insight	Demand exists, but current growth is value-eroding. Scaling without fixing margins risks diluting overall portfolio profitability.
Objective	Restore economic efficiency by addressing pricing, discounting, and cost leakage before pursuing additional growth.
Recommendations	• Review pricing and promotional depth to identify margin leakage. • Optimize cost-to-serve (logistics, returns, supplier terms) to support profitable growth. • Pause aggressive expansion until margin performance stabilizes. • Test price elasticity carefully to rebalance growth and profitability.

Strategic Recommendations

4. Deprioritize — Minimize Exposure to Value-Destructive Segments	
Product	Sofas
Insight	This product generates negative value across both volume and margin dimensions and should not compete for scarce resources without a compelling recovery case.
Objective	Limit capital allocation and management attention due to weak value creation.
Recommendations	• Reduce inventory exposure and discretionary marketing spend. • Consider SKU rationalization, repositioning, or phased exit. • Attempt turnaround only if a clearly defined, economically viable strategy exists (e.g., premium repositioning or cost restructuring).